



A 20-Gb/s charge-steering equalizer utilizing highly-efficient charge-steering linear equalizer

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ABSTRACT

The need for high-speed low-power receivers has grown due to the large increase in data rates and Internet-of-Things (IoT) applications. Equalizer-based receivers contain mainly two types of equalizers; Continuous-Time-Linear-Equalizers (CTLE) and non-linear equalizers implemented as Decision-Feedback-Equalizers (DFE). Many solutions have been introduced in the literature to enhance the power consumption of DFEs by using time interleaving or charge-steering. Still, the CTLE power consumption remains considerable a big portion of the overall power of the equalizer-based receivers. The CTLE function cannot be neglected as it equalizes the pre-cursors. This paper presents a low-power charge-steering based receiver that consists of a 1/8-rate charge-steering DFE integrated with a newly proposed Charge-Steering-Linear-Equalizer (CSLE). The proposed CSLE is explained qualitatively and quantitatively. The entire receiver chain consumes 2.9-mW running from a 0.9-V supply and can compensate for 14-dB channel loss at Nyquist-rate when processing 20-Gb/s data.

1. Introduction

With the extensive usage of Internet-of-Things (IoT) devices in everyday life, from sensors to autonomous cars, the need to process and share data at high speeds increases especially with the emergence of 5G technology [1]. Most of these devices need to operate while consuming low power to run for long periods without recharging. As the data rates increase, the frequency-dependent loss introduced by the channel increases, as a result, the demand for equalization at the receiver side to compensate for inter-symbol interference (ISI) has increased. Most equalizer-based receivers tend to use Continuous-Time-Linear-Equalizers (CTLE) which provides peaking at the desired frequency with low complexity. However, it has a major disadvantage of amplifying crosstalk and noise-induced by the channel which affects the signal to noise ratio (SNR). To mitigate this, a Decision-Feedback-Equalizer (DFE) is used after the CTLE, as it has the advantage of equalizing the signal without amplifying noise or crosstalk. DFEs cannot equalize pre-cursors though and CTLE is still needed.

Charge-Steering (CS) has been used for designing low-power circuits that are capable of operating at high frequencies. Its main idea is replacing the tail current source with a charge source and the load with switched capacitors [2] as shown in Fig. 1. The small-signal gain of the

CS amplifier shown in Fig. 1 can be expressed as:

$$A_v \approx \frac{2C_T}{C_D} \quad (1)$$

C_D and C_T are explained in Table A.4 in Appendix A. A CTLE is simply an amplifier with degeneration impedance that provides a peaking at Nyquist frequency. It acts as a high-pass filter to compensate for the low-pass effect of the channel. Its transfer function is given by (2).

$$H(s) = \frac{g_m(R_D // SC_P)}{(1 + g_m(0.5R_s // 2SC_s))} \quad (2)$$

g_m , C_P and R_D , R_s and C_s are explained in Table A.4 in Appendix. A DFE is a non-linear equalizer that is used to compensate for the limited performance of the CTLE. The DFE acts as a filter that uses feedback with a certain transfer function to estimate the response of the channel. The DFE input is taken from the output of the CTLE and its output is subtracted from its input after passing through a sample and hold circuit, thus the effect of the post-cursors ISI is diminished as shown in Fig. 2.

Receivers that target to achieve high equalization tend to use both CTLE and DFE as each one nearly compensates for the other's drawback. However, with the increase in data rates, the power consumed by both

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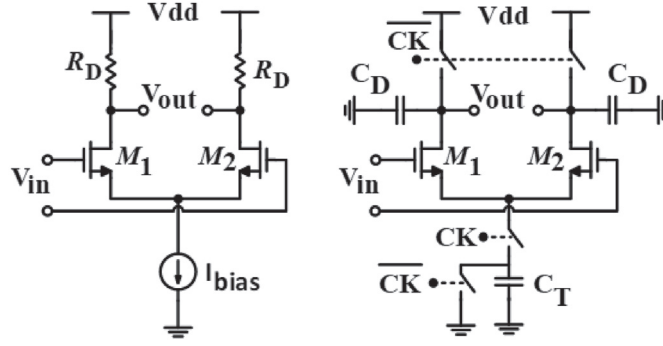


Fig. 1. Transformation from current-steering to charge-steering [2].

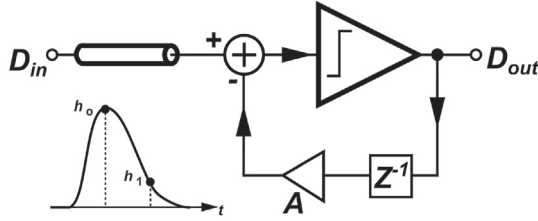


Fig. 2. Simplified DFE block diagram.

equalizers increases dramatically. A lot of techniques have been published to improve the power consumption of DFEs such as using half-rate or quarter-rate DFEs and also by using charge steering summers and latches [3]. At data rates of multiple Gb/s, the feedback timing becomes more critical even for half-rate and quarter-rate DFEs. On the other hand, CTLEs consume 18% of the overall analog power consumption in the receiver/equalizer, for example, the CTLE consumes 4.7-mW out of 26-mW at 10.8 Gb/s [4]. Since CTLEs are continuously operating, time interleaving is impossible and there is always a path from the supply to the ground which increases power consumption. Also, in Ref. [5] an active inductor CTLE is introduced which consumes 7.57-mW at 20 Gb/s achieving a FOM of 12.6-fJ/bit/dB. Several alternatives were introduced in the literature to replace the CTLE or reduce its power consumption. For example, a discrete-time linear-equalizer (DTLE) was introduced in Ref. [6] where the output is valid for half the clock cycle only. Operating two charge-based CTLEs in different phases was proposed in Ref. [6]. Both designs [6,7] suffered from output invalidity through the entire clock cycle causing tough timing constraints on the following stage.

This paper presents a CS-based equalizer. A newly proposed architecture for the CTLE named Charge-Steering-Linear-Equalizer (CSLE) is introduced that allows achieving an ultra-low power-consumption compared to conventional CTLEs. Also, a 1/8-rate DFE utilizing the CS technique for both the summers and the slicers is used. This allows high-speed operation with low power. This paper is organized as follows. Section 2 describes the circuits and the derived equations of the proposed receiver. Section 3 presents the equalizer simulation results. Finally, the conclusions are given in Section 4.

2. Proposed receiver

The proposed receiver is shown in Fig. 3. The CSLE uses a complementary clock at 20-GHz, it is then followed by eight sample and hold circuits of the DFE. The 1/8-rate 1-tap DFE uses eight clocks with a 50-ps delay between each one, and a frequency of 2.5 GHz. The clocks can be generated from a Clock-Data-Recovery (CDR) or from Injection-Locked-Ring-Oscillator (ILRO) [8] which proved to be a low power solution for clock generation.

2.1. CSLE

The proposed CSLE depends on charge steering instead of current steering, thus it is expected to be more power-efficient. It operates in 2 phases; equalization (phase 1) and amplification (phase 2). The operation of the CSLE is shown in Fig. 4. Dummy switches for charge sharing mitigation are added but not shown in the figure for simplicity. In phase 1, the CSLE operates as a normal linear equalizer where it amplifies the input signal. The gain of the CSLE similar to the CTLE where the frequency response is determined by the load impedance and the degeneration impedance. To understand the gain and operation of the CSLE, each phase is discussed separately and thoroughly. In phase 1, if the on-resistance of M5 is neglected then the capacitance seen at node X can be given by (3)

$$C_T = C_{p1} + C_{T1} \quad (3)$$

C_T represents the total capacitance that will store the charges drained from M3 which operates in the triode region providing a resistance named R_L during phase 1. C_{p1} is the summation of all capacitance at node X and C_{T1} is the tail capacitor in the CSLE.

The current flowing into that tail capacitor can be represented by (5) as defined in Ref. [5]. Deriving the gain of phase 1 starts with defining the current flowing in this phase.

$$\int_0^{\Delta T} I_{CT} dt = C_T \int_0^{V_{CM}-V_{TH}-\Delta V} dV \quad (4)$$

$$I_{CT} = \frac{1}{2} \mu_n C_{ox} \frac{2W}{L} (V_{CM} - V - V_{TH})^2 \quad (5)$$

$$\Delta T \approx \frac{C_T}{\mu_n C_{ox} \frac{W}{L}} \frac{(V_{CM} - V_{TH} - \Delta V)}{(V_{CM} - V_{TH}) \Delta V} \quad (6)$$

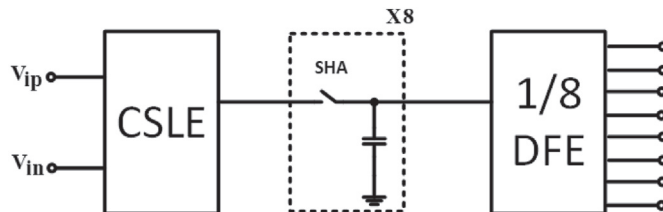


Fig. 3. Proposed receiver.

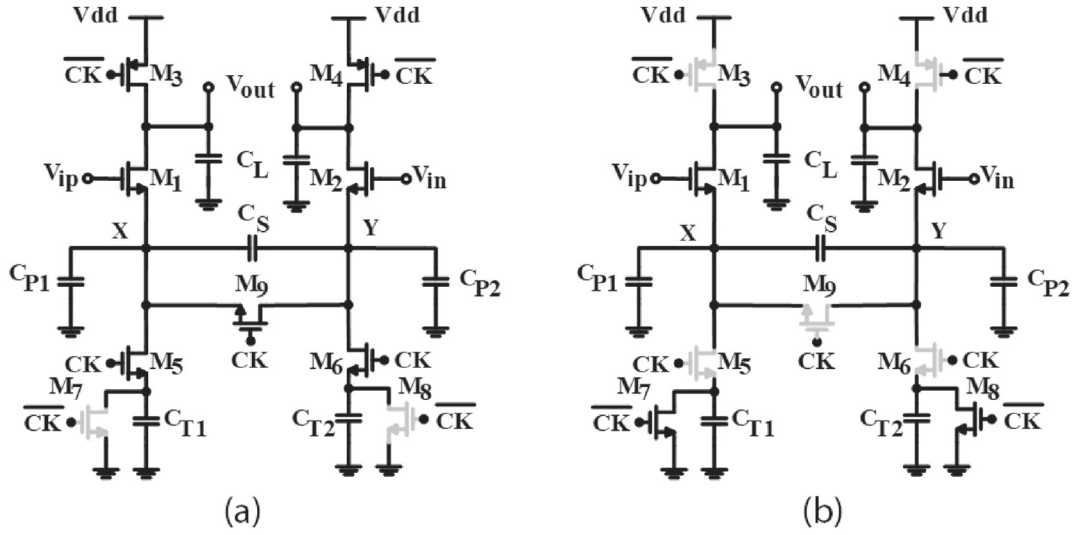


Fig. 4. CSLE operation: a) Phase 1 b) Phase 2.

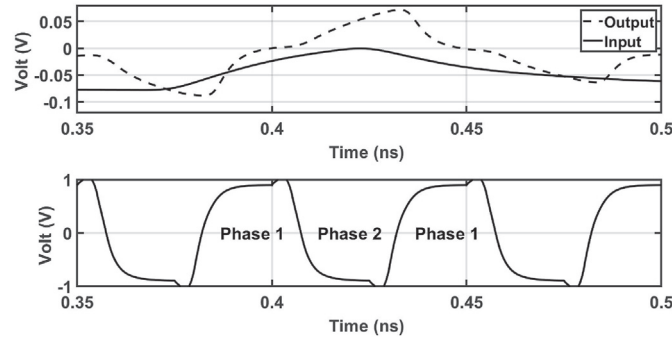


Fig. 5. CSLE transient simulation.

ΔT is the time needed to make the nodes X and Y reach $V_{CM} - V_{TH}$. Assuming that the region of operation of the diff-pair is saturation, then the current following inside the diff-pair can be driven as in (7). Reformulating I_{avg} drained through the diff-pair by substituting ΔT in (7) with expression in (5), thus (8) is obtained.

$$I_{avg} = \frac{C_T(V_{CM} - V_{TH} - \Delta V)}{\Delta T} \quad (7)$$

$$I_{avg} = \mu_n C_{ox} \frac{W}{L} (V_{CM} - V_{TH}) \Delta V \quad (8)$$

The average transconductance provided by the input pair is shown in (9), then by using (8) the final expression of $g_{m,avg}$ (10) is obtained.

$$g_{m,avg} = \frac{2I_D}{V_{GS} - V_{TH}} \approx \frac{I_{avg}}{(V_{GS} - V_{TH})_{avg}} \quad (9)$$

$$g_{m,avg} \approx \frac{\mu_n C_{ox} \frac{W}{L} (V_{CM} - V_{TH}) \Delta V}{(V_{CM} - V_{TH} + \Delta V)/2} \quad (10)$$

The output voltage and voltage gain in phase 1 can be driven in (11) and (12), respectively.

$$V_{out} \approx \frac{g_{m,avg} V_{in} Z_L}{1 + g_{m,avg} Z_s} \quad (11)$$

$$A_{v,ph1} \approx \frac{g_{m,avg} Z_L}{1 + g_{m,avg} Z_s} \quad (12)$$

During phase 1 the node X and Y are shorted through R_s . The CSLE is acting as a common source amplifier with a degeneration impedance Z_s . Applying the half-circuit model to obtain the impedance seen through

the degeneration and substituting in (12) again. Finally, the gain for phase 1 is derived in (13).

$$A_{v,ph1} \approx \frac{g_{m,avg}(R_L // SC_L)}{1 + g_{m,avg} \left(\frac{R_s}{2} // SC_{T,total} \right)} \quad (13)$$

$$C_{T,total} = C_{p1} + 2C_s + C_{T1} \quad (14)$$

The peak gain for phase 1 is given by:

$$A_{v,ph1} \approx g_{m,avg} R_L \quad (15)$$

In the transition from phase 2 to phase 1, the capacitors C_{T1}/C_{T2} are under reset after releasing them from the reset condition at the start of phase 1. Both capacitors start charge sharing with C_{p1}/C_{p2} , respectively. As both nodes X/Y are shorted through M9 in phase 1, this can be considered as a soft reset for nodes X/Y. It is important to reset both nodes before starting a new equalization phase. At the beginning of phase 2 the output capacitors (C_L) acts as a charge source starting with an initial condition taken from phase 1. The initial condition depends on the input swing applied to the CSLE in phase 1. If the swing is low or moderate then the output voltage at the start of phase 2 will be determined through (11). The two nodes X/Y “source of input pairs” are now shorted by only C_s . If the voltage at nodes X/Y is far away from $V_{CM} - V_{TH}$ at the end of phase 1, then the CSLE in phase 2 can be seen as Return-Zero (RZ) CS amplifier. The load capacitance will act as a charge source and the tail capacitance (C_{p1}) absorbs the charges till the end of phase 2. This allows the equalizer to amplify only the wanted signal without amplifying low frequency noise and distortions outside

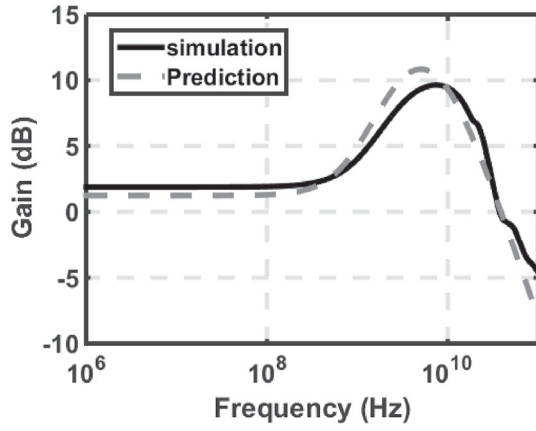


Fig. 6. Predicted vs simulated transfer function.

the bandwidth of the CSLE Peak. For phase 2, the transfer function is given by:

$$A_{vph2} = \frac{2 \left(C_{p1} + \frac{C_s * C_{p1}}{C_s + C_{p1}} \right)}{C_L} \quad (16)$$

Fig. 5 shows the transient simulation for the CSLE input and output in the 2 phases. It can be seen that in phase 1, the CSLE equalizes the input signal, while in phase 2, it acts as a CS amplifier giving the equalized signal the gain shown in (16).

2.1.1. CSLE simulation results

It can be seen from (16) that phase 2 acts as an RZ CS amplifier at all frequencies. The proposed CSLE consumes 183 μ W when operating at full-rate (20 GHz). Fig. 6 shows the predicted transfer function. This is obtained by applying the transfer functions in (13) and (16) in MATLAB where the values of R and C are obtained from Cadence. The simulated one is obtained from Periodic-AC (PAC) analysis where the load capacitance is equal to 10.8-fF (one sample and hold in addition to the input capacitance of the summer). The results show a reasonable agreement with the equation driven and the approximations made in (4) to (7). To capture the gain of the CSLE across one complete clock

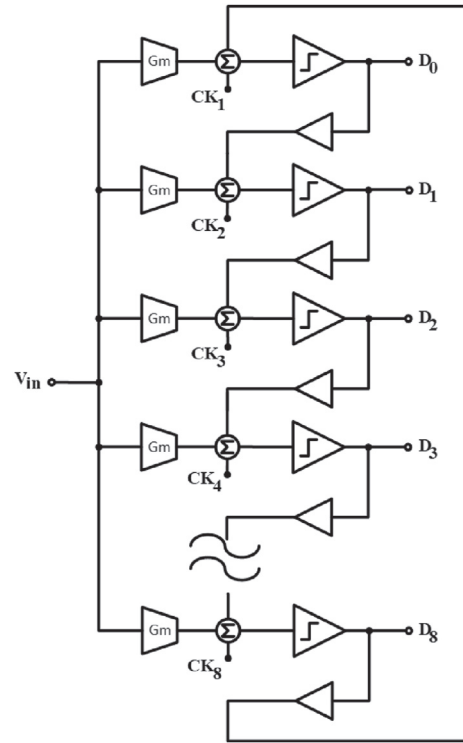
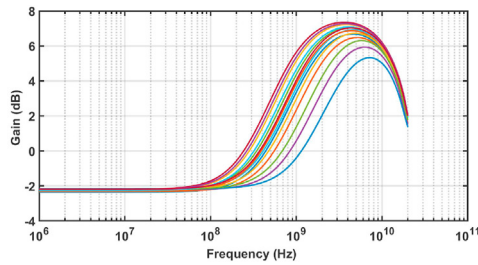


Fig. 8. 1/8 DFE block diagram.

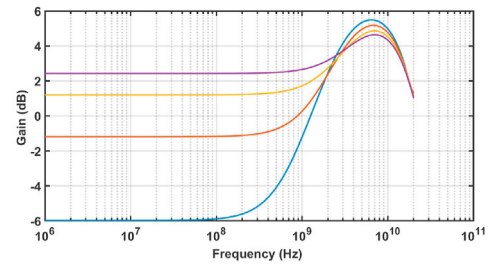
cycle Periodic-Steady-State (PSS) analysis was used along Periodic-AC (PAC).

The CSLE transfer function can be tuned to change the zero location by varying CS using switches as shown in Fig. 7(a). The peaking can be tuned by either using multiple M9 transistors and switches to change the overall sizing or varying the gate voltage of M9 instead of using CLK. The latter option is applied, and Fig. 7(b) shows the tuning results.

Table 1 shows the performance comparison of the CSLE and prior topologies of linear equalizers. The best peaking is achieved in Ref. [4], but it operates at a lower data rate and has a high-power consumption



(a) Varying Cs value



(b) Varying M9 gate voltage

Fig. 7. CSLE tuning results.

Table 1
CSLE Performance comparison.

	JSSC20 [4]	MEJ20 [5]	MWCAS15 [6]	JSSC15 [7]	This Work
Technology (nm)	65	28	40	45	65
Data rate (Gb/s)	1.62–10.8	20	8	25	20
Architecture	Cherry Hooper CTLE	Active inductor CTLE	DTLE	Inductor based CTLE	CSLE
Peak gain (dB)	20	5.5–9.5	11	8	8
Power (mW)	4.7	7.57	0.19	2.4	0.18/0.63 ^a
Power efficiency (fJ/bit)	435.2	378.5	24	96	9/31.5 ^a

^a Including the power of CSLE clock buffers.

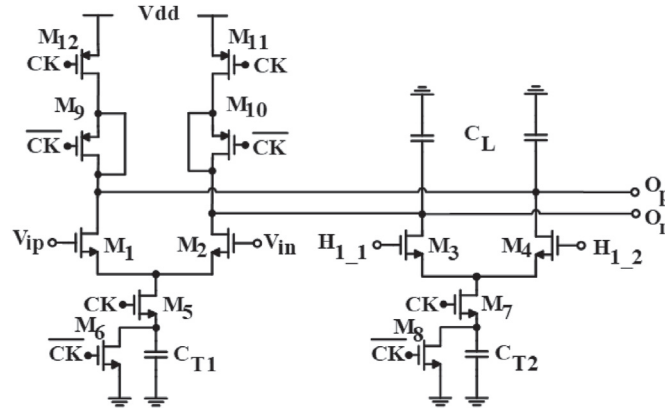


Fig. 9. Charge Steering Summer [2].

meanwhile, in Ref. [5], an active inductor solution was used to avoid the usage of passive inductors still the power consumption is high compared to others. In Ref. [6] a low power consumption was achieved, but it operates at a much lower data rate than other counterparts in Table 1. Finally [7], achieved a lower power than [4,5], while operating at a high data rate, however, it uses a passive inductor that occupies most of the receiver area nearly 60% of the entire receiver area. The proposed CSLE consumes very low power with considerable peak gain and high data rate thus achieving ultra-low power efficiency compared to prior work.

The proposed CSLE is superior in performance when compared to CTLE or DTLE counterparts. The CTLE is continuously draining current from the supply to maintain its output validity through the entire cycle. On the other hand, the DTLE is operating in a single-phase where the output is not valid for the entire cycle. The proposed CSLE combines both the advantages of the CTLE and the DTLE where it does not drain continuous current from the supply and its output remains valid throughout the entire clock cycle. The CSLE also utilize the CS concept to be more power-efficient than other counterparts. The integration of the CS circuits usually requires more attention than regular circuitry. The issue comes from the impedance seen from the proceeding stage and the sampling events that can happen at the output capacitance. As long as the sampling and hold capacitor of the proceeding stage is relatively small in comparison to the load capacitance of the CS circuit no buffering is required. Buffers are power-hungry especially when they are operating at high data-rates, thus avoiding them is one of the main design goals that must be achieved. Also in the proposed equalizer, the CSLE is using full-rate clock which requires stringent requirements

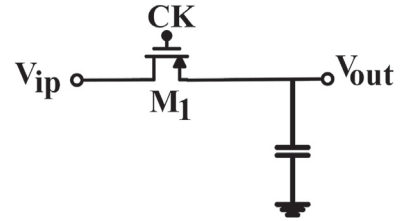


Fig. 10. Sample and hold.

on the clock buffer especially at 20-GHz, and if higher data rates are required, then full-rate approach can't be used and it will be more efficient to apply half-rate or even quarter-rate architectures to the CSLE.

2.2. 1/8-rate DFE, slicer

To test the compatibility of the CSLE in an equalizer chain, the CSLE is added before the equalizer proposed in Ref. [3] which consists of a 1/8-rate DFE architecture shown in Fig. 8. It consists of eight single-tap DFEs, where CS summers, with embedded CS Gm, shown in Fig. 9 and slicers are cascaded. The output of each slicer is fed to the tap of the next summer. The timing constraint must be satisfied in less than one Unit-Interval (UI) of 50-ps as per (17). The 1/8 architecture allows both the summer and slicer to work at 2.5-GHz instead of 20-GHz which simplifies the design and decreases their delay and power consumption.

$$t_{cq} = t_{setup} + t_{feedback} \leq UI \quad (17)$$

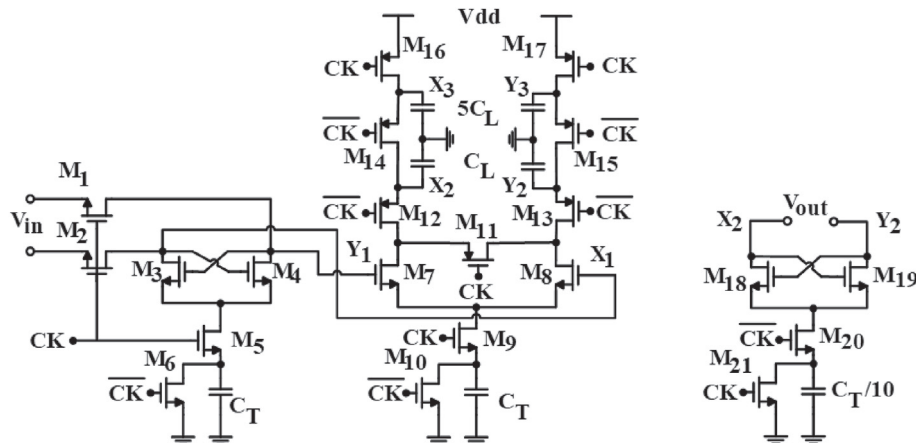


Fig. 11. Non-return-to-zero charge steering Flip-Flip (slicer) [2].

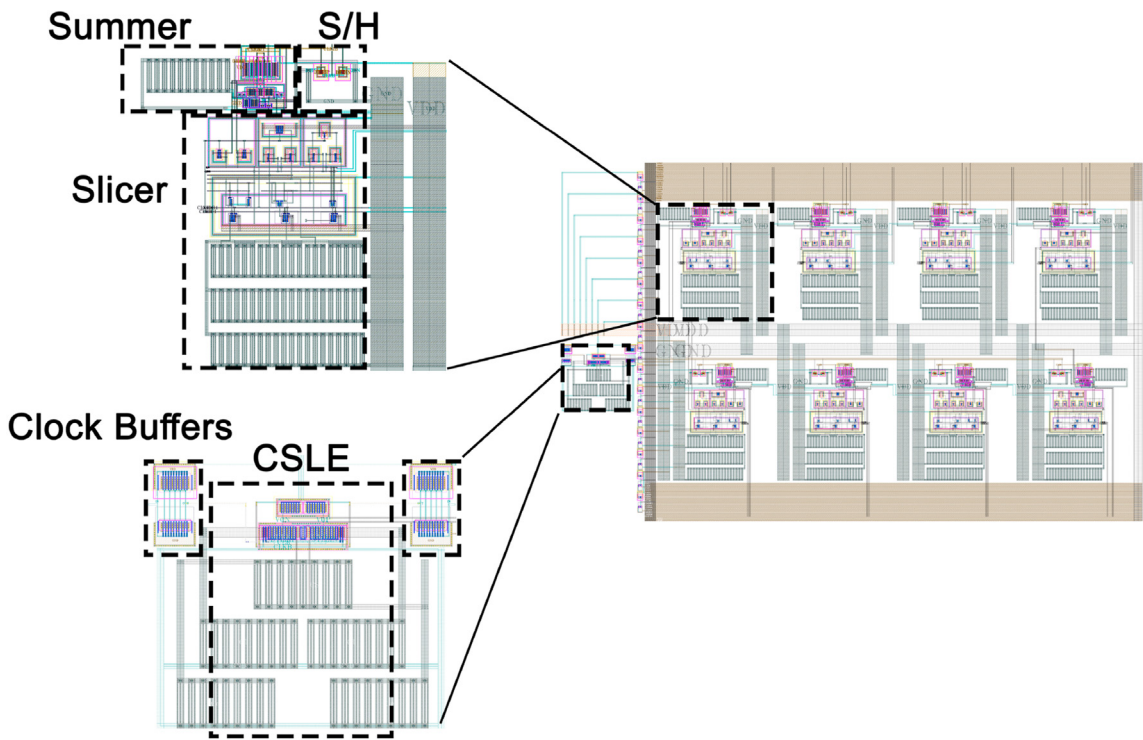


Fig. 12. Receiver layout.

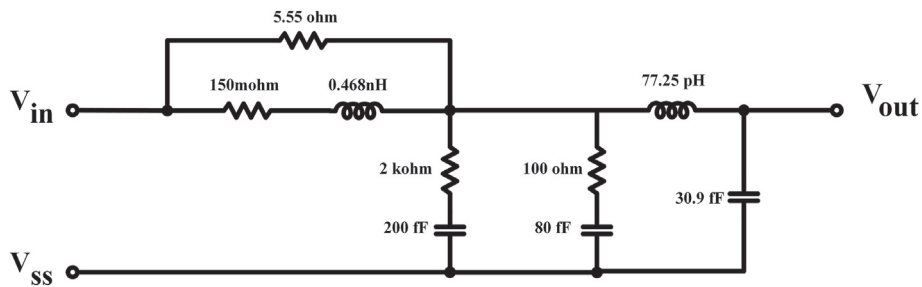


Fig. 13. One-inch FR4 channel model [9].

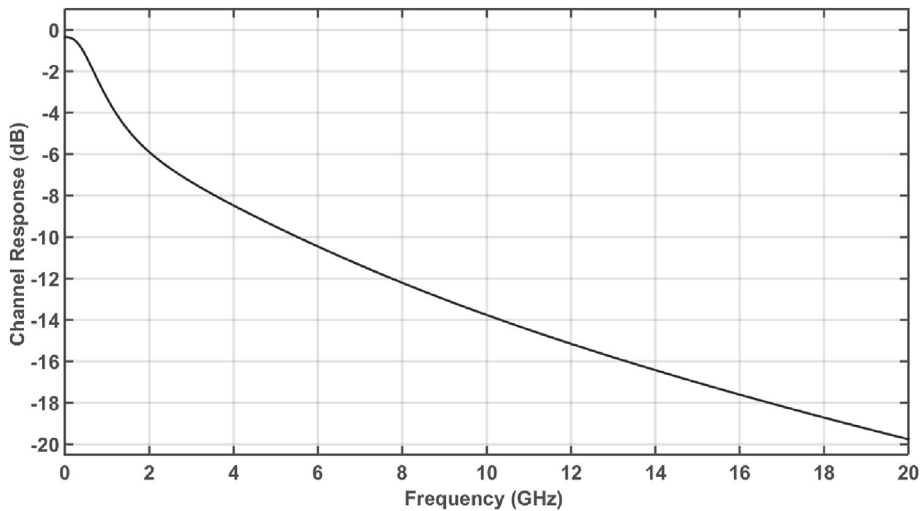


Fig. 14. Channel frequency response for 14-inch.

Where t_{cq} is the worst clock to output delay, t_{setup} is the time needed for the input to be valid from the previous DFE stage before the clock transition and $t_{feedback}$ is the time needed for the summer output (slicer

input) to settle. A careful layout is mandatory to satisfy the timing requirement in (17). The usage of higher metals is highly recommended to decrease the parasitic capacitance for the routing between the slicer

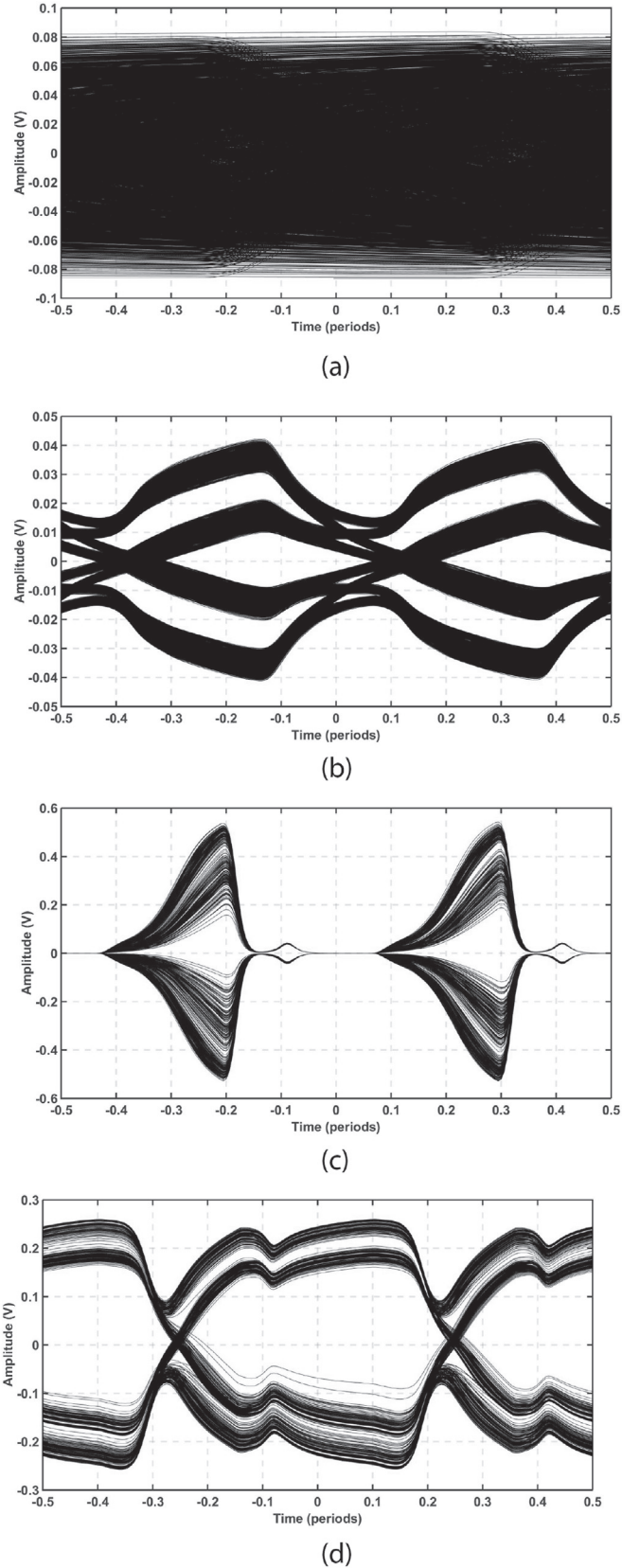


Fig. 15. Eye diagrams: (a) CSLE input. (b) CSLE output. (c) Summer output. (d) Slicer output.

and the summer.

The integration between the CSLE and the Sample and Hold (S/H) before the summer must be done with care to avoid using a buffer between them. Implementing a buffer at that frequency is a power-hungry block that will consume a lot of extra power. One of the main problems in the integration was the Non-Return-to-Zero (NRZ) latch used as a S/H and the regeneration circuit [3]. It creates an extra loading to the CSLE that causes charge injection to the CSLE in its reset phase. To avoid this problem, a normal S/H, shown in Fig. 10, is used where the capacitor is sized so that to avoid extra load on the CSLE.

Another integration problem is between the summer and the slicer after removing the regeneration circuit in the S/H, the output common-mode and swing from the summer decreased. Two solutions can solve this problem; using PMOS input pairs in the slicer or resizing the summer tail capacitor to achieve a reasonable common-mode inside the slicer. The first solution was avoided as extra power will be required to achieve the same transconductance of the NMOS input pair, this makes the second solution much more desired.

The slicer and delay are implemented as Non-Return-to-Zero (NRZ) Charge-Steering (CS) Flip-Flop (FF) as it proved to be better in terms of speed and power consumption when compared to CMOS and CML FFs [7]. However, CS FF output is not valid for the entire clock cycle due to the reset phase in CS which requires an extra circuit or special clocking to connect to digital blocks. The CS FF shown in Fig. 11 solves the previous problem as its output is valid for the entire clock cycle using only complementary clocks [3].

The CS FF consists of two latches; the first latch is a conventional NRZ latch and the other one is a modification to the RZ latch presented in Ref. [7]. The operation of the CS FF is clearly explained in Ref. [3]. The CS FF consumes only 20 A when operating at 2.5 Gb/s. The gain of the first stage of the NRZ latch inside the slicer shown in Fig. 11 depends on the input signal swing from the summer. The gain for the first stage of the NRZ is similar to the regular CS amplifier gain in (2) meanwhile the large-signal output voltage is given by (18).

$$V_{out} = \frac{C_T}{C_D} \left(\frac{(V_{CM} - V_{TH})^2 + \frac{3}{4} V_{in}^2}{V_{CM} - V_{TH} + \frac{V_{in}}{2}} \right) \quad (18)$$

The analysis for the large signal gain is shown in Ref. [6] and from (18) the gain can be seen as the small-signal gain (2) and from (18) the gain or the output voltage depends also on the input common-mode and input pair thresholds. The input signal from the summer can be considered as a small signal which means the first stage of the NRZ will satisfy (2). Thus, the output voltage dependency will be limited to the ratio between the tail capacitor, the load capacitor, and the input signal swing. This is more practical to have a mature design, that should be able to withstand any Process-Voltage-Temperature (PVT) variations.

3. Equalizer simulation results

The receiver layout is shown in Fig. 12. It occupies an area of $251 \mu\text{m} \times 152 \mu\text{m}$; nearly 85% of this area is occupied by the MOM capacitors used for the charge-steering circuits, and the CSLE occupies an area of $25 \mu\text{m} \times 31 \mu\text{m}$ which is nearly 2% of the receiver total area.

One-inch for the channel model in Ref. [9] is shown in Fig. 13. The channel model can be cascaded to stand for a specific length for the channel. The channel model for 14-inch has 14 dB loss at 10 GHz as shown in Fig. 14. The channel model in Ref. [9] doesn't include any sources to introduce cross-talk noise among different channels, it can only be considered only as a good representation for the FR4 channel loss across different frequencies.

The receiver is designed using 65-nm CMOS technology. The operation of the circuit is verified using random ideal inputs of 180-mv peak-to-peak swing and rise and fall times of 10 ps. A 14-inch of the FR4 channel model in Ref. [9] was used in the post-layout simulation to represent the channel losses. A 0.5-nH inductor and 25-mΩ resistance were added to the ideal supply and ground to model the 0.1-nH bump and 0.4-nH PCB trace. The post-layout-simulated eye diagrams of the CSLE, DFE, and slicer are shown in Fig. 15. The CSLE was able to provide a vertical eye-opening of 15 mv and horizontal eye-opening of 75.2% of the UI, while the vertical eye-opening for the receiver output was 282 mv and horizontal eye-opening was 95% of the UI. Worst case corner results are shown in Table 2. The supply variation used is $\pm 5\%$ and the temperature range is from -40°C to 85°C . The results show that high temperature causes degradation in the vertical eye-opening, but this can be fixed by injecting extra charges in the slicer (by using

Table 2
Worst case corners.

Transistors Corner	VDD (V)	Temperature ($^\circ\text{C}$)	Vertical eye opening (mV)	Horizontal eye opening
SS	0.855	85	131	93
FF	0.945	-40	525	96
FS	0.855	85	223	95
SF	0.855	85	298	96
FS	0.945	-40	287	84
SF	0.945	-40	407	96

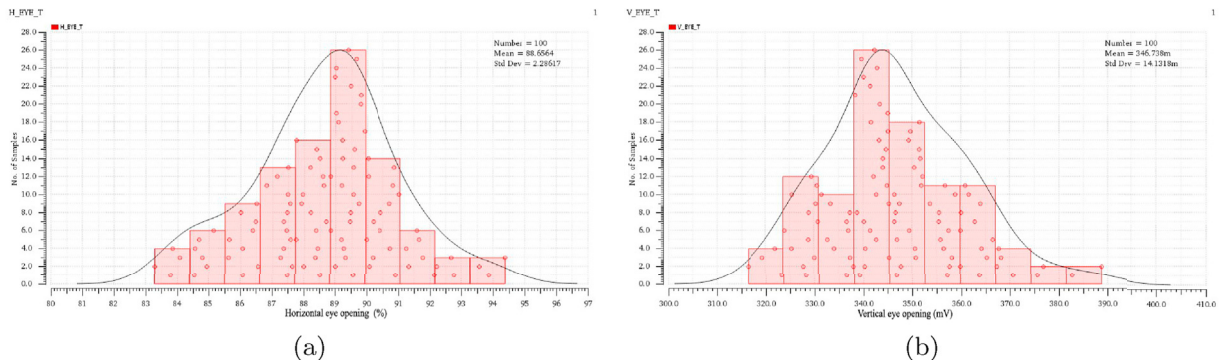


Fig. 16. Monte Carlo results: (a) Horizontal eye opening (b) Vertical eye opening.

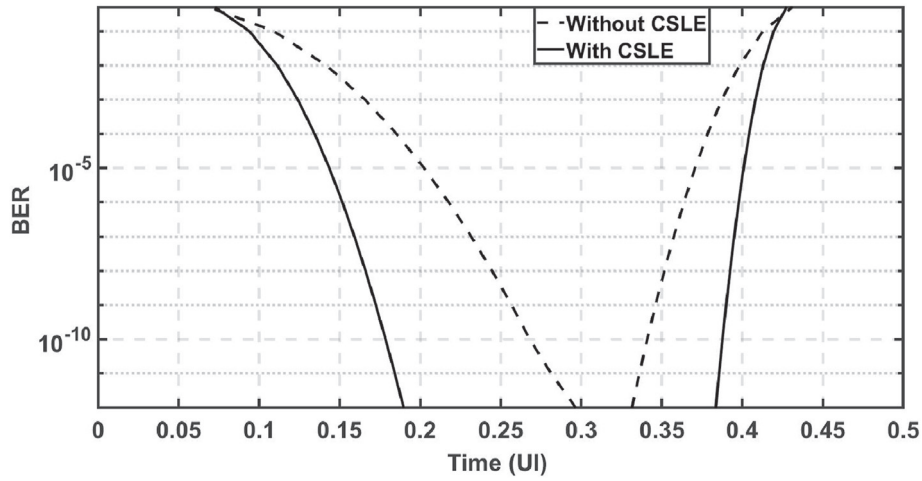


Fig. 17. Bathtub curve.

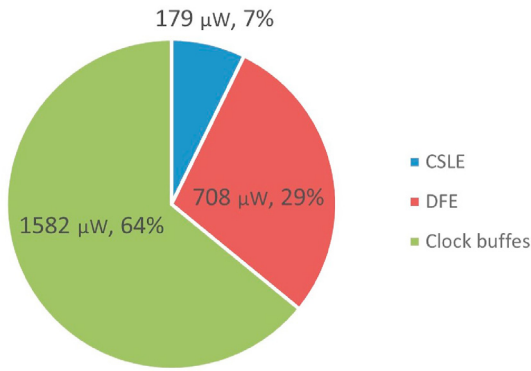


Fig. 18. Power breakdown.

capacitor bank for example). Monte Carlo simulation is performed to get the offset voltage which will be needed for bit error rate (BER) calculation. The results for vertical and horizontal eye openings histograms are shown in Fig. 16. The results show standard deviations of 2.28% in horizontal eye opening and 14-mV in vertical eye opening. Using the results from Monte Carlo and transient noise, BER is calculated and a bathtub curve is plotted using MATLAB in Fig. 17. This shows that the proposed equalizer was able to achieve a BER of 10^{-12} for 0.2 UI. The total power consumption of the receiver is 2.9-mW resulting in power efficiency of 145-fJ/bit. Both the CSLE and the slicer have the advan-

tage of the validity of the data for the entire clock cycle. This will facilitate the integration with the demultiplexer. A sufficient eye-opening is required at the input of the demultiplexer for correct detection. The equalizer can be considered successful if the output data from the slicer is XORed with input data to the channel and the output from the XOR is zero. The entire equalizer is using a charge-steering technique that is proven efficient while operating at high data rate.

Fig. 18 shows the power breakdown of the receiver, it is seen then more than half the power consumed by the receiver is due to the clock buffers while the equalization part (CSLE and DFE) consumes 36% of the total power consumption with only 7% for the CSLE. Table 3 presents the comparison results with prior work. The table shows that the proposed receiver has better power efficiency than previous publications operating at similar data rates. This is attributed to the use of the newly proposed CSLE and the use of optimized charge-steering circuits.

4. Conclusion

This work presents a 20-Gb/s CS equalizer with a newly proposed CSLE and 1/8-rate DFE with one discrete tap. The entire equalizer is utilizing the CS concept to boost the equalizer power-efficiency. The newly proposed CSLE was proven to be superior when compared to its counterparts. Designed in 65-nm CMOS technology, the proposed equalizer can compensate for a channel loss up to 14 dB. The power consumption of the equalizer is 2.9 mW from a 0.9-V supply running at

Table 3
Performance summary and comparison.

	TVLSI 16 [12]	JSSC 16 [10]	TCAS II 20 [11]	JSSC 20 [4]	TCAS II 20 [13]	This work
Technology (nm)	65	65	28 LP	65	55	65
Data rate (Gb/s)	21	20	18	1.62–10.8	56	20
Architecture	CTLE 1-tap DFE	IIR DFE	3 stage CTLE 1-tap DFE	CTLE 2-tap DFE	2-tap FIR ^d 2-tap DFE	CSLE 1-tap DFE
Max loss (dB)	−14.9	−5.8	−32	−34	−13	−14
Supply (V)	1.2	–	1.2	1.18/1.1/1	1	0.9
Power (mW)	34.2	14.1	58.8 ^b	37.2	4.89 ^c	2.9
Power efficiency (fJ/bit)	1630	710	3267	3400	87	145
FoM ^a (fJ/bit/dB)	110	122	102	100	6.72	10.36
Area (mm ²)	0.027	0.027	0.07315 ^e	0.174	0.002679	0.0382

^a FoM = (power efficiency)/(channel loss @ Nyquist frequency).

^b Calculated power for DFE and CTLE only.

^c Area of the receiver only.

^d Ideal FFE with 2 taps was used.

^e Including estimated power of the clock buffers.

20 Gb/s with a power efficiency of 145 fJ/bit. The equalizer occupies a total area of $251 \mu\text{m} \times 152 \mu\text{m}$.

Authorship contributions

M.A.Saif: Conception and design of study, acquisition of data, analysis and/or interpretation of data, Drafting the manuscript, revising the manuscript critically for important intellectual content, Approval of the version of the manuscript to be published, K.M.Hassan: Conception and design of study, acquisition of data, analysis and/or interpretation of data, Drafting the manuscript, revising the manuscript critically for important intellectual content, Approval of the version of

the manuscript to be published, S.A.Ibrahim: Conception and design of study, acquisition of data, analysis and/or interpretation of data, Drafting the manuscript, revising the manuscript critically for important intellectual content, Approval of the version of the manuscript to be published,

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Terms description

Table A.4

Terms description.

Term	Description	Equation
C_D	Total capacitance at the output nodes	(1)
C_T	The tail capacitor	
gm	The transconductance of the input pair	(2)
R_D	Load resistor	
C_p	Load capacitor	
R_S	Degeneration resistor	
C_S	Degeneration capacitor	
V_{CM}	Output common-mode voltage	(6)
ΔV	Differential voltage applied on the diff-pair	
$\frac{w}{L}$	Input pair size	
Z_L	Load impedance	(12)
Z_S	Degeneration impedance	

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